

HOLDEN AGDM, AB, Canada

WMO: 712270

Lat: **53.1850N**Lon: **112.2461W**Elev: **2257**StdP: **13.54**Time Zone: **-7.00 (NAM)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.7	-18.3	-29.9	1.1	-23.5	-24.4	1.5	-18.1	26.4	23.9	23.5	20.3	7.1	220	0.714

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	83.5	63.0	80.1	61.6	77.1	60.3	66.3	78.9	64.1	75.9	62.2	73.1	9.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.8	89.8	71.6	59.7	83.1	69.0	57.7	77.2	66.9	32.2	78.8	30.5	76.1	29.0	73.2	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.9	21.4	18.9	DB	-32.8	89.7	5.5	3.7	-36.7	92.3	-40.0	94.5	-43.1	96.6	-47.1	99.2	(4)
(5)			WB	-32.9	70.4	5.5	2.4	-36.9	72.1	-40.1	73.5	-43.2	74.9	-47.3	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	11.2	10.9	22.3	38.0	50.3	57.9	62.6	60.1	51.2	38.9	23.0	11.7		
	DBStd	21.85	15.33	14.03	13.26	9.43	7.65	5.33	5.04	5.72	7.95	8.94	12.47	13.95		
	HDD50	6058	1204	1094	859	368	90	7	0	3	79	356	811	1187		
	HDD65	10407	1669	1514	1324	810	455	221	107	169	416	809	1261	1652		
	CDD50	1184	0	0	0	8	100	243	391	315	115	12	0	0		
	CDD65	58	0	0	0	0	1	7	33	15	2	0	0	0		
	CDH74	1403	0	0	0	2	93	177	554	404	169	4	0	0		
	CDH80	310	0	0	0	0	12	24	142	87	45	0	0	0		
	WSAvg	9.3	9.9	9.2	9.7	10.4	10.2	9.2	8.2	7.9	8.7	9.5	9.2	9.2		
	PrecAvg	15.1	0.7	0.4	0.6	1.3	1.6	2.5	2.9	2.3	1.2	0.7	0.5	0.6		
(15) Precipitation	PrecMax	19.8	2.3	1.0	1.4	2.7	2.6	4.4	6.2	4.3	3.2	1.2	1.3	1.3		
	PrecMin	7.6	0.1	0.1	0.2	0.2	0.2	0.6	0.9	0.4	0.2	0.1	0.0	0.1		
	PrecStd	2.8	0.5	0.3	0.4	0.7	0.6	1.1	1.3	0.9	0.8	0.4	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	41.3	54.2	71.5	81.7	82.8	88.8	86.9	86.0	72.6	55.5	43.6		
		MCWB	35.9	35.4	42.8	50.7	56.7	62.0	66.7	63.0	60.3	54.1	44.9	37.4		
	2%	DB	38.7	37.7	46.2	65.6	76.3	78.9	84.0	82.1	78.9	63.3	46.7	38.7		
		MCWB	33.5	33.3	38.9	48.2	54.5	60.0	65.5	62.4	57.5	49.6	39.6	34.1		
	5%	DB	36.0	34.7	41.7	59.1	72.0	75.1	80.0	78.6	73.1	57.8	41.8	34.9		
		MCWB	32.4	31.4	36.2	44.6	53.0	58.6	64.0	61.9	55.7	46.2	36.2	31.4		
	10%	DB	32.4	31.5	38.1	54.4	67.5	71.3	76.4	74.7	67.0	53.1	38.1	31.2		
		MCWB	29.6	29.0	33.5	42.2	50.8	56.7	62.2	60.5	53.2	43.4	33.8	28.5		
	0.4%	WB	36.6	36.3	44.6	51.8	58.4	65.4	70.2	68.2	61.3	54.7	45.0	37.7		
		MCDB	40.6	40.6	52.7	70.4	76.0	78.8	82.4	78.6	80.1	70.0	54.2	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.5	33.8	39.6	48.5	55.9	62.4	67.3	65.2	58.8	50.9	40.2	34.4		
		MCDB	38.1	37.0	45.8	64.3	72.4	73.8	79.7	77.4	76.1	61.7	45.8	38.3		
	5%	WB	32.4	31.6	36.5	45.3	54.0	60.2	65.3	62.9	56.6	47.4	36.8	31.6		
		MCDB	35.6	34.6	41.3	57.2	69.1	71.1	77.5	74.8	70.9	56.1	40.8	34.6		
	10%	WB	29.6	28.8	34.0	42.9	52.0	58.4	63.3	61.1	54.1	44.5	34.2	28.4		
		MCDB	32.9	31.7	37.4	53.0	65.0	69.1	74.1	72.7	65.6	51.7	37.7	31.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	19.2	18.5	21.4	24.7	22.0	22.9	24.0	24.1	21.2	17.0	17.2		
		MCDBR	18.3	21.2	20.8	31.2	32.2	28.8	30.1	31.8	35.3	29.7	20.9	19.5		
		MCWBR	15.1	17.8	14.5	16.3	14.5	12.9	14.3	15.5	17.3	16.9	14.5	15.8		
		MCDBR	18.6	20.5	19.8	28.6	29.0	24.2	27.3	28.1	32.3	28.3	19.8	19.2		
	5% WB	MCWBR	15.9	17.3	14.1	15.8	13.7	12.1	13.8	14.6	16.3	16.4	14.4	16.1		
(40) Clear-Sky Solar Irradiance	taub		0.258	0.264	0.304	0.331	0.348	0.365	0.359	0.372	0.312	0.279	0.253	0.245		
	taud		2.301	2.357	2.343	2.361	2.367	2.348	2.389	2.329	2.472	2.539	2.446	2.290		
	Ebn at Noon		233	273	281	286	286	281	280	269	277	266	240	216		
	Edh at Noon		18	24	30	34	36	37	35	35	27	20	16	15		
(44) All-Sky Solar Radiation	RadAvg		322	654	1131	1489	1827	1911	1976	1619	1125	656	321	238		
	RadStd		34	51	86	140	127	127	99	115	119	80	35	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (85 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page